

SUSOBHAN GHOSH

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EDUCATION	Ph.D. in Computer Science Harvard University	08/2020 - 05/2026
	• Advisor: Prof. Susan A. Murphy • Research area: Bayesian Reinforcement Learning (RL) for digital health interventions	
	M.S. and B.Tech (Hons.) in Computer Science IIIT Hyderabad	2015 - 2019
	• GPA: 9.29/10.00, Rank: 1 (Gold Medalist).	
EXPERIENCE	Applied Scientist II Amazon	07/2026 - Present
	• Working on ranking and personalization for Amazon.com promotions and deals	
	Applied Scientist II Intern Amazon	06/2025 - 12/2025
	• Designing RL algorithms for Amazon's Supply Chain Optimization to bridge the sim-to-real gap by incorporating hard to model real-time constraints	
	Doctoral Researcher Harvard University	09/2019 - 05/2026
	• In collaboration with University of Michigan, served as the RL-tech lead and a core member in a multi-year research project called MiWaves. MiWaves is a digital intervention to help reduce cannabis use among emerging adults (age 18-25). • Designed an RL algorithm called reBandit which utilizes mobile sensor and self-reported data to help deliver digital interventions in any healthcare study. reBandit was utilized in the clinical trial for MiWaves with 122 participants, which resulted in a 77% participant engagement rate. • Published peer-reviewed scientific articles at AAAI, IJCAI, IAAI, AAMAS, ISCA, Philosophical Transactions of Royal Society, and Machine Learning Journal. • Invited Guest speaker at venues like SAA, SCT, INFORMS and ENAR, and courses at Harvard like Statistical RL. • Served as a peer-concentration advisor for 2 undergraduate students	
	Research Engineer Singapore Management University	09/2019 - 12/2020
	• In collaboration with Google and Wildlife Conservation Trust India, led the design of a multi-year multidisciplinary research project to help reduce human-wildlife conflicts. • Developed an AI-based prediction tool to determine areas with high risk of human-wildlife conflicts (published at AAAI). As of 2024, tool currently used by WCT India at Bramhapuri Forest Division in Chandrapur, Maharashtra, India.	
	Graduate Research Assistant IIIT Hyderabad	09/2018 - 07/2019
	• Developed an autonomous broker named VidyutVanika V2 for trading renewable energy in smart grid markets, which placed 2nd in 2018 Power Trading Agent Competition (PowerTAC). • Published the tariff, wholesale and balancing electricity market autonomous algorithms of VidyutVanika V2 at AAAI.	
	Research Intern TCS Innovation Labs, Hyderabad, India	05/2018 - 07/2018
	• Designed trading algorithms for real-time load balancing in power markets.	
	Research Intern TCS Innovation Labs, Hyderabad, India	05/2017 - 07/2017
	• Led the development of VidyutVanika V1, an autonomous broker – which finished 4th in 2017 PowerTAC competition.	
	Google Summer of Code LibreOffice	05/2016 - 09/2016
	• Designed and developed 7 panels and 3 decks for sidebars for the LibreOffice suite (available in LibreOffice v5.3+). • LibreOffice releases with my sidebars are being used by over 200 million users (as of 2025).	

PUBLICATIONS

1. **Susobhan Ghosh***, Bhanu T. Gulapalli*, Daiqi Gao, Asim Gazi, Anna Trella, Ziping Xu, Kelly Zhang, Susan A. Murphy. “Reproducible workflow for online AI in digital health” *Philosophical Transactions of The Royal Society A*, 2026.
2. **Susobhan Ghosh***, Anna Trella*, Erin Bonar, Lara Coughlin, Finale Doshi-Velez, Yongyi Guo, Pei-Yao Hung, Inbal Nahum-Shani, Vivek Shetty, Maureen Walton, Iris Yan, Kelly Zhang, Susan Murphy. “Effective monitoring of online AI decision-making algorithms in just-in-time adaptive interventions”. *npj Digital Medicine*, 2026.
3. Maureen Walton, Inbal Nahum-Shani, Maya Campbell, Devin C Tomlinson, Autumn Rae Florimbio, **Susobhan Ghosh**, Yongyi Guo, Pei-Yao Hung, Mark W Newman, Jeremy J Lin, Tianchen Qian, John Dziak, Huijie Pan, Kelly W Zhang, Lauren Zimmermann, Erin E Bonar, Susan A. Murphy, Lara N Coughlin. “Mobile intervention for emerging adults with regular cannabis use: a micro-randomized trial”. *The Lancet Regional Health–Americas* 61, 2026
4. Asim Gazi, **Susobhan Ghosh**, Yongyi Guo, Daiqi Gao, Ziping Xu, Inbal Nahum-Shani, Susan A. Murphy. “Key Concepts in Online Learning and Decision Making for Personalizing Just-in-Time Adaptive Interventions (pJITAI)” *Under Review*, 2026
5. Asim H. Gazi, Daiqi Gao, **Susobhan Ghosh**, Ziping Xu, Anna Trella, Predrag Klasnja, Susan A. Murphy. “Digital Twins for Just-in-Time Adaptive Interventions (JITAI-Twins): A Framework for Optimizing and Continually Improving JITAI” *Journal of Medical Internet Research*, 2026.
6. **Susobhan Ghosh**. “Deployable Online Reinforcement Learning Algorithms for Use-inspired Research in Digital Health.” *PhD Thesis, Harvard University*, 2026.
7. **Susobhan Ghosh**, Pei-Yao Hung, Lara N. Coughlin, Erin E. Bonar, Yongyi Guo, Inbal Nahum-Shani, Maureen Walton, Mark W. Newman, and Susan A. Murphy. “It felt more real: Investigating the User Experience of the MiWaves Personalizing JITAI Pilot Study.” *EAI International Conference on Pervasive Computing Technologies for Healthcare*, 2025.
8. **Susobhan Ghosh**, Yongyi Guo, Pei-Yao Hung, Lara Coughlin, Erin Bonar, Inbal Nahum-Shani, Maureen Walton, Susan Murphy. reBandit: Random Effects based Online RL algorithm for Reducing Cannabis Use. *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
9. **Susobhan Ghosh***, Raphael Kim*, Prasad Chhabria, Raaz Dwivedi, Predrag Klasnja, Peng Liao, Kelly Zhang, Susan Murphy. “Did we personalize? Assessing personalization by an online reinforcement learning algorithm using resampling”. *Machine Learning Journal*, 2024.
10. Lara Coughlin, Maya Campbell, Tiffany Wheeler, Chavez Rodriguez, Autumn Rae Florimbio, **Susobhan Ghosh**, Yongyi Guo, Pei-Yao Hung, Mark Newman, Huijie Pan, Kelly Zhang, Lauren Zimmermann, Erin Bonar, Maureen Walton, Susan Murphy, Inbal Nahum-Shani. “A mobile health intervention for emerging adults with regular cannabis use: A micro-randomized pilot trial design protocol”. *Contemporary Clinical Trials*, 2024.
11. **Susobhan Ghosh**, Yongyi Guo, Pei-Yao Hung, Lara Coughlin, Erin Bonar, Inbal Nahum-Shani, Maureen Walton, and Susan Murphy. “MiWaves Reinforcement Learning Algorithm” *arXiv preprint arXiv:2408.15076* (2024).
12. Srivatsan Krishnan, Amir Yazdanbakhsh, Shvetank Prakash, Jason Jabbour, Ikechukwu Uchendu, **Susobhan Ghosh**, Behzad Boroujerdian, Daniel Richins, Devashree Tripathy, Aleksandra Faust, Vijay Janapa Reddi. “ArchGym: An Open-Source Gymnasium for Machine Learning Assisted Architecture Design”. *International Symposium on Computer Architecture (ISCA)*, 2023
13. Arpita Biswas, Jackson Killian, Paula Diaz, **Susobhan Ghosh**, Milind Tambe. Fairness for Workers Who Pull the Arms: An Index Based Policy for Allocation of Restless Bandit Tasks. *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023.

15. Aditya Mate, Arpita Biswas, Christoph Siebenbrunner, **Susobhan Ghosh**, Milind Tambe. Efficient Algorithms for Finite Horizon and Streaming Restless Multi-Armed Bandit Problems. *International Conference on Autonomous Agents and Multiagent Systems (AA-MAS)*, 2022.
16. **Susobhan Ghosh**, Pradeep Varakantham, Aniket Bhatkhande, Tamanna Ahmad, Anish Andheria, Wenjun Li, Aparna Taneja, Divy Thakkar, Milind Tambe. Facilitating human-wildlife cohabitation through conflict prediction. *AAAI Conference on Artificial Intelligence (AAAI)*, 2022.
17. Haipeng Chen, **Susobhan Ghosh**, Gregory Fan, Nikhil Behari, Arpita Biswas, Mollie Williams, Nancy E. Oriol, Milind Tambe. Using Public Data to Predict Demand for Mobile Health Clinics. *AAAI Conference on Artificial Intelligence (AAAI)*, 2022.
18. **Susobhan Ghosh**, Sujit Gujar, Praveen Paruchuri, Easwar Subramanian, Sanjay P. Bhat. Bidding in Smart Grid PDAs: Theory, Analysis and Strategy. *AAAI Conference on Artificial Intelligence (AAAI)*, 2020.
19. Sanjay Chandekar, Bala Suraj Pedasingu, **Susobhan Ghosh**, Easwar Subramanian, Sanjay Bhat, Praveen Paruchuri, Sujit Gujar. VidyutVanika: AI-Based Autonomous Broker for Smart Grids: From Theory to Practice. *Energy Sustainability through Retail Electricity Markets, Applied Innovation and Technology Management Book Series*, 2023.
20. **Susobhan Ghosh**, Easwar Subramanian, Sanjay P. Bhat, Sujit Gujar, Praveen Paruchuri. VidyutVanika: A Reinforcement Learning Based Broker Agent for a Power Trading Competition. *AAAI Conference on Artificial Intelligence (AAAI)*, 2019.
21. **Susobhan Ghosh**. Learning Strategies for Power Trading in Smart Grids. *MS Thesis, IIIT Hyderabad*, 2019.
22. **Susobhan Ghosh**, Kritika Prakash, Sanjay Chandekar, Easwar Subramanian, Sanjay P. Bhat, Sujit Gujar, Praveen Paruchuri. VidyutVanika: An Autonomous Broker Agent for Smart Grid Environment. *Policy, Awareness, Sustainability and Systems (PASS) Workshop*, 2019.

TALKS & LECTURES

- **JITAI-Twins: A Digital Twin Framework for Reinforcement Learning Algorithms and Just-in-Time Adaptive Interventions**, INFORMS 2025 (Invited Talk)
- **Deploying RL algorithms for digital interventions**, ENAR 2025 (Invited Talk)
- **reBandit: Personalizing Treatment Delivery for Reducing Cannabis use**, INFORMS 2024 (Invited Talk)
- **reBandit: Random Effects based Online RL algorithm for Reducing Cannabis Use**, IJCAI 2024
- **MiWaves: AI-driven Digital Health Interventions to help reduce cannabis use**, SAA 2024
- **RL in real life: mixed effects models and reBandit**, STAT234 (Spring 2024, 2025): Statistical Reinforcement Learning at Harvard University (Guest Invited Lecture)
- **MiWaves: Using AI to help reduce cannabis use among emerging adults**, d3C Think Tank Jan 2024 (Invited)
- **MiWaves: Developing a 2nd Generation JITAI to reduce cannabis use among emerging adults**, d3C Think Tank May 2023 (Invited)
- **MiWaves**, d3C Think Tank Oct 2022 (Invited)
- **VidyutVanika: A Reinforcement Learning Based Broker Agent for a Power Trading Competition**, AAAI 2019
- **Review of Sidebar and its functionality**, LibreOffice Conference 2016 (Invited Talk)

TEACHING	• Teaching Assistant , Classics in Computer Science, Harvard University	2022
	• Teaching Assistant , Information Security, IIIT Hyderabad	2019, 2018
	• Teaching Assistant , Introduction to Game Theory, IIIT Hyderabad	2017
	• Teaching Assistant , Computer Networks, IIIT Hyderabad	2017
	• Teaching Assistant , Introduction to Software Engineering, IIIT Hyderabad	2016
	• Teaching Assistant , Network Management Workshop, IIT Kharagpur	2015
	• Teaching Assistant , Network Management Workshop, Nettech Pvt. Ltd.	2014
AWARDS AND HONORS	• Siebel Scholarship Award , Siebel Foundation	2025
	• Best Reviewer Award , AISTATS 2025	2025
	• Gold Medalist , IIIT Hyderabad	2019
	• AAAI Student Scholarship , AAAI	2019
	• Microsoft Research Travel Grant	2019
	• Dean's Merit List , IIIT Hyderabad	2016, 2017, 2018
	• Central Sector Scheme of Scholarship (CSSS) , India	2013
SKILLS	• Prime Minister Scholarship Scheme (PMSS) , India	2013
	Languages: English, Hindi, Bengali. Programming Languages: Python, C++, C, Java. Others: Personalization, Multi-armed bandits, Machine Learning, Bayesian Reinforcement Learning, Sequential Decision-Making, Regression Models, Data Science, Digital Interventions, Mobile Health, Mixed-effects Models, Autonomous Agents, Human-Computer Interaction (HCI), Sim-to-real, Deep Learning, Deep Reinforcement Learning, Robotics	
COMMUNITY SERVICES	Reviewer for: <i>NeurIPS-26, AISTATS-26, NYRL 2025, NeurIPS-25, AISTATS-25, NeurIPS-24, RLC-24</i> Open Source Contributor: <i>LibreOffice</i>	